

Experiment ‘p_6_2_a_swap_first_last_seq’ Results

June 5, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_6_2_a_swap_first_last_seq

Natural language statement: null

Method signature: p_6_2_a_swap_first_last_seq(inputs: seq<int>) returns (result: seq<int>)

Ensures

- `|result| == |inputs|`
- `result[0] == inputs[|inputs| - 1]`
- `result[|inputs| - 1] == inputs[0]`
- `forall i :: 1 <= i < |inputs| - 1 ==> result[i] == inputs[i]`

Requires

- `|inputs| >= 2`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_6_2_a_swap_first_last_seq(inputs: seq<int>) returns (result: seq<int>)`

The method should respect the following contract:

`requires |inputs| >= 2, ensures |result| == |inputs|, ensures result[0] == inputs[|inputs| - 1], ensures result[|inputs| - 1] == inputs[0], ensures forall i :: 1 <= i < |inputs| - 1 ==> result[i] == inputs[i]`

Produce and show only the Dafny body of this method, including the curly braces that surround it. Do not show the signature nor contract. You must not use 'assume' nor 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DnNgQKIvVPRNb2Xr0xNaq3ZWPTZjc

Model: gpt-5.5-2026-04-23

Created at: 1780661482

Finish reason: stop

Usage: 380 tokens in, and 561 tokens out

```
{
  result := inputs[0 := inputs[|inputs| - 1]][|inputs| - 1 := inputs[0]];
}
```

Final Program

```
method p_6_2_a_swap_first_last_seq(inputs: seq<int>) returns (result: seq<int>)
  requires |inputs| >= 2
  ensures |result| == |inputs|
  ensures result[0] == inputs[|inputs| - 1]
  ensures result[|inputs| - 1] == inputs[0]
  ensures forall i :: 1 <= i < |inputs| - 1 ==> result[i] == inputs[i]
{
  result := inputs[0 := inputs[|inputs| - 1]][|inputs| - 1 := inputs[0]];
}
```

Total Token Usage

Input tokens: 380

Output tokens: 561

Reasoning tokens: 512

Sum of ‘total tokens’: 941

Experiment Timings

Overall Experiment started at 1780661481568, ended at 1780661510835, lasting 29267ms (29.27 seconds)

Iteration #1 started at 1780661481569, ended at 1780661510835, lasting 29266ms (29.27 seconds)